Analysis notes on cycling data

Station Identifiers were not stable over the analysis period. Several stations were assigned multiple IDs while retaining the same station name. For analytical purposes, station names were used to represent physical stations, while station IDs were retained as historical identifiers

Analysis Findings So Far

1. Ride Duration by Hour

Average ride duration differs noticeably between members and casual riders. Members generally take shorter, steadier, and more consistent rides throughout the day. Casual riders take longer rides on average, with duration increasing substantially during the daytime and reaching its highest levels roughly between 9am and 5pm.

This suggests that casual riders may use the service differently from members, potentially for longer leisure or recreational trips, while members appear to have a more consistent riding pattern.

2. Total Rides by Hour

Total ride volume is relatively similar for members and casual riders during the late-night/early-morning hours, particularly around 11pm–4am. The groups begin to separate during the morning, with noticeable peaks around 9am and 5pm.

The 9am and 5pm peaks are consistent with typical commuting periods. The stronger member volume around these peaks is particularly relevant when comparing member and casual behaviour.

3. Rideable Type Over Time

In 2024, classic bikes and electric bikes appear relatively evenly matched in total ride volume. During 2025 and into early 2026, electric bikes become much more prominent, with the difference especially visible during the summer and autumn of 2025.

This is an area worth investigating further. A useful next step is to break the electric-bike increase down by member_casual to determine whether the change is being driven mainly by members, casual riders, or both.

4. Ride Duration by Month

Casual riders consistently have longer average ride durations than members. The difference is most obvious during the warmer months and becomes much smaller during the winter months.

This seasonal pattern strengthens the idea that casual riding is more associated with longer, potentially leisure-oriented trips, while member riding is comparatively consistent throughout the year.

5. Total Rides by Month and Rider Type

Members account for more rides than casual riders across the period analysed. The gap between the two groups is relatively consistent over time, although the absolute volume of rides for both groups rises and falls with the seasons.

This establishes an important baseline: members generate more total ride volume, but total rides alone do not fully describe how the two groups use the service. Ride duration, time of day, seasonality, rideable type, and station behaviour provide additional distinctions.

6. Start Station Patterns

The highest-volume start stations differ between member and casual riders. Among the top stations for casual rides are Shedd Aquarium, Field Museum, Navy Pier, DuSable Harbor, Buckingham Fountain, Adler Planetarium, and McCormick Place. This concentration around major attractions and waterfront/recreational locations suggests that casual riding is strongly associated with central tourist and leisure areas.

The highest-volume member stations include Kingsbury St & Kinzie St, Clinton St & Washington Blvd, Clinton St & Madison St, Clark St & Elm St, and Wells St & Concord Ln. These locations appear more closely associated with areas used by regular riders, although this should be treated as an observation rather than a confirmed explanation.

The station maps are therefore more useful when used to compare member and casual behaviour rather than simply showing where the most rides occur. A particularly useful future visual is a map where bubble size represents the percentage of rides from casual riders at each station, while applying a minimum ride-volume threshold.

7. Overall Patterns Emerging

• Members take more rides overall.

• Members generally take shorter and more consistent rides.

• Member ride volume is particularly strong around typical commuting periods.

• Casual riders take longer rides on average.

• The casual-member duration gap widens during warmer months and narrows during winter.

• Casual rides are more concentrated around major attractions, waterfront areas, and other leisure-oriented locations.

• Electric bikes became substantially more prominent during 2025, creating a useful follow-up question about which rider group drove the increase.

8. Questions Worth Investigating Next

• Is the 2025 increase in electric-bike usage driven mainly by casual riders, members, or both?

• Do casual riders disproportionately start and/or end trips near tourist attractions?

• Are casual riders more likely to take longer rides because of where they start, where they end, or both?

• How do member and casual ride patterns differ by day of week as well as hour of day?

• Do the busiest member and casual stations have different peak hours?

• Which stations have the highest percentage of casual rides after applying a minimum ride-volume threshold?

• How does the member-versus-casual difference change across seasons and years?

Note: These findings are observations from the current Power BI analysis. They should be treated as patterns or hypotheses to investigate further rather than causal conclusions.